Day 2 .
Networking Devices

1. What Are Networking Devices?

Networking devices connect computers, servers, and other devices to communicate and share data across a network.

Important networking devices:

- Router
- Switch
- Hub
- Bridge
- Repeater
- Wireless Access Point (WAP)
- Firewall
- Modem
- Network Interface Card (NIC)
- Proxy server
- Load balancer

2. Types of Networking Devices

1. Router

- Connects different networks.
- Forwards packets using IP addresses.
- Selects the best available route to a destination.
- Commonly operates at **OSI Layer 3 (Network)**.

Example: Connects your home network to the internet.

Cybersecurity use: Can provide basic packet filtering and network segmentation.



1.Router

2. Switch

- Connects devices within a LAN.
- Forwards Ethernet frames using MAC addresses.
- Learns MAC addresses and stores them in a MAC address table.
- Primarily operates at **Layer 2 (Data Link)**.

Example: Connects computers in a college computer lab.

Cybersecurity use: VLANs can separate departments or isolate network traffic.



2.switch

3. Hub

- Connects multiple Ethernet devices.
- Sends incoming signals to all other ports.
- Does not learn MAC addresses or intelligently filter traffic.
- Operates at **Layer 1 (Physical)**.

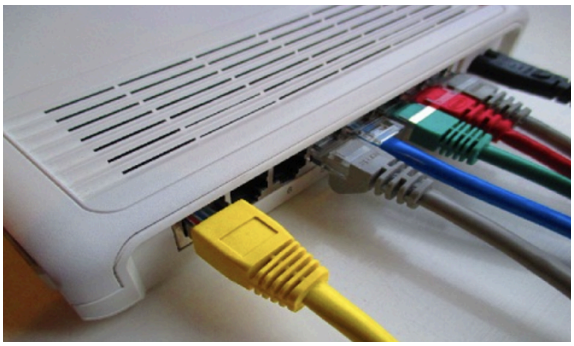
Remember: Hubs are largely obsolete and have been replaced by switches.



4. Bridge

- Connects or divides LAN segments.
- Forwards or filters frames using MAC addresses.
- Operates at **Layer 2 (Data Link)**.

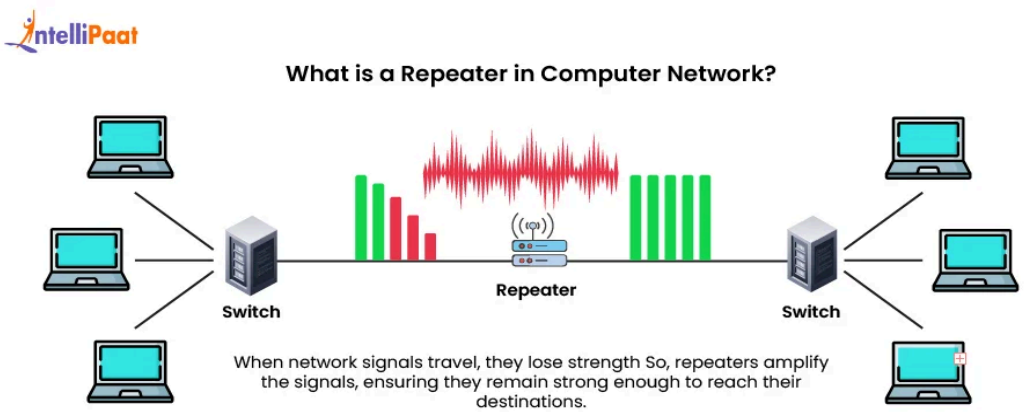
Remember: A switch is essentially a multiport bridge



5. Repeater

- Regenerates or retimes signals to extend their transmission distance.
- Does not route traffic or filter frames.
- Operates at **Layer 1 (Physical)**.

Example: Extends a network signal across a longer distance.



6. Wireless Access Point (WAP)

- Allows wireless devices to connect to a wired LAN using Wi-Fi.
- Bridges wireless clients to the network.
- Commonly associated with **Layer 2**.

Example: A college Wi-Fi access point connects laptops and smartphones to the campus network.

Cybersecurity tip: Use WPA2-AES or WPA3 and strong authentication to protect wireless networks.

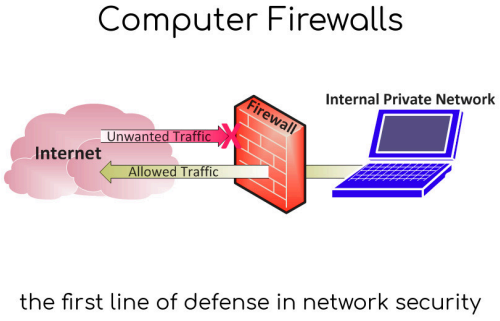


7. Firewall

- Monitors and controls network traffic based on security rules.
- Can allow or block traffic based on IP addresses, ports, protocols, and connection state.
- Advanced firewalls may inspect application traffic.

Example: Blocks unauthorized incoming connections to a server.

Cybersecurity use: Helps prevent unauthorized network access.



8. Modem

- Converts signals to enable communication over a particular transmission medium.
- Connects a network to an ISP through technologies such as cable or DSL.

Example: A cable modem provides an internet connection through a cable network.

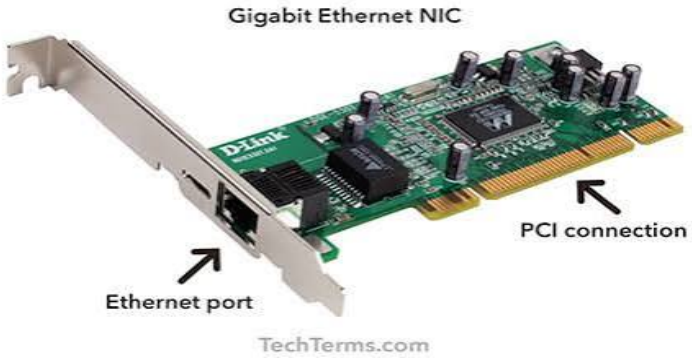
Remember: A modem and a router perform different functions, although some home devices combine both.



9. Network Interface Card (NIC)

- Provides a device with a network connection.
- Can support Ethernet or Wi-Fi.
- Typically has a MAC address for Layer 2 communication.

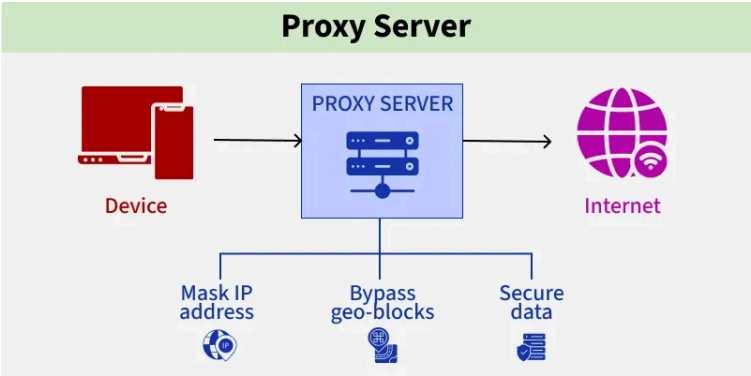
Example: An Ethernet adapter connects a desktop computer to a switch.



10. Proxy Server

- Acts as an intermediary between a client and another server.
- A forward proxy handles requests on behalf of clients.
- A reverse proxy receives requests on behalf of backend servers.

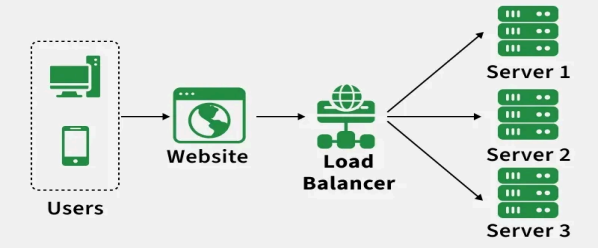
Cybersecurity use: Can filter web requests, enforce access policies, and hide internal client addresses from external destinations.



11. Load Balancer

- Distributes incoming requests across multiple servers.
- Helps improve performance, scalability, and availability.
- Can operate at Layer 4 or Layer 7, depending on its design.

Example: Distributes website traffic across several web servers.



3. Important Comparison Table

Device	Main Function	OSI Layer	Key Identifier
Hub	Sends signals to all ports	Layer 1	None
Repeater	Regenerates signals	Layer 1	None
Switch	Forwards frames within a LAN	Layer 2	MAC address
Bridge	Connects LAN segments	Layer 2	MAC address
Router	Connects different networks	Layer 3	IP address
Wireless AP	Connects Wi-Fi clients to a LAN	Primarily Layer 2	MAC address
Firewall	Filters traffic according to rules	Depends on type	IP, ports, state, and more
Modem	Converts signals for transmission	Depends on technology	Signal/modulation information
NIC	Provides network connectivity	Layers 1 and 2 functions	MAC address
Load balancer	Distributes network requests	Layer 4 or 7	IP/ports or application data

Exam tip: Learn the primary OSI layer for each device. Some devices perform functions across multiple layers.

4. Collision Domain vs. Broadcast Domain

These concepts are important for networking exams.

Collision Domain

A network area in which devices could experience frame collisions when sharing a medium.

- A hub places connected devices in the same collision domain.
- Each switch port creates a separate collision domain.

Broadcast Domain

A network area in which a Layer 2 broadcast can be received.

- Routers separate broadcast domains by default.
- VLANs create separate logical broadcast domains.
- Switches normally forward broadcasts within the same VLAN.

Remember:

- Switch → Separates collision domains.
- Router → Separates broadcast domains.
- VLAN → Separates logical broadcast domains.

5. Managed vs. Unmanaged Switches

Managed Switch	Unmanaged Switch
Can be configured	Usually works without configuration
Supports features such as VLANs	Usually lacks advanced management features
Provides monitoring and control options	Provides basic connectivity
Suitable for enterprise networks	Suitable for simple home networks

Cybersecurity tip: Managed switches can support VLANs, port security, and monitoring features that help protect enterprise networks.

6. Quick Revision Notes

Memorize these points before moving to the next topic:

- **Router:** Connects networks; IP addresses.
- **Switch:** Connects LAN devices; MAC addresses.
- **Hub:** Sends signals to all ports.
- **Bridge:** Connects LAN segments.
- **Repeater:** Regenerates signals.
- **WAP:** Connects wireless clients to a LAN.
- **Firewall:** Filters traffic for security.
- **Modem:** Converts signals for transmission.
- **NIC:** Provides network connectivity.
- **Proxy:** Acts as an intermediary.
- **Load balancer:** Distributes requests across servers.
- **VLAN:** Creates logical network separation.